



Write or stick your student
number here

2026 HIGHER SCHOOL CERTIFICATE TRIAL EXAMINATION

Mathematics Extension 2

General instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Stick or write your Student Number at the top of this page and on each answer booklet

Total marks: 100

Section I – 10 marks (pages 2–4)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 5–13)

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

- 1 Given $z_1 = 4e^{\frac{i\pi}{3}}$ and $z_2 = 2e^{\frac{i\pi}{6}}$, which of the following is the complex number $\frac{z_1}{z_2}$ expressed in Cartesian form?

- A. $2(1 + i\sqrt{3})$
- B. $2(\sqrt{3} + i)$
- C. $1 + i\sqrt{3}$
- D. $\sqrt{3} + i$

- 2 Consider the statement:

‘If a triangle is equilateral, then it is isosceles’.

Which of the following is the converse of that statement?

- A. If a triangle is isosceles, then it is equilateral.
- B. A triangle is equilateral if and only if it is isosceles.
- C. If a triangle is not equilateral, then it is not isosceles.
- D. If a triangle is not isosceles, then it is not equilateral.

- 3 The three cube roots of a complex number w are z_1 , z_2 and z_3 .

If $z_1 = 2i$, what are the other two roots?

- A. $1 - i\sqrt{3}$ and $-1 - i\sqrt{3}$
- B. $-1 - i\sqrt{3}$ and $-1 + i\sqrt{3}$
- C. $\sqrt{3} - i$ and $-\sqrt{3} - i$
- D. $-\sqrt{3} - i$ and $-\sqrt{3} + i$

- 4 Consider the statement:

$$\forall x \in \mathbb{Z}, \ln|x| > 0$$

Which of the following is the negation of this statement?

- A. $\exists x \in \mathbb{Z}$, such that $\ln|x| > 0$
 - B. $\exists x \in \mathbb{Z}$, such that $\ln|x| \leq 0$
 - C. $\forall x \in \mathbb{Z}$, $\ln|x| \leq 0$
 - D. $\ln|x| > 0 \Rightarrow x \in \mathbb{Z}$
- 5 A particle's velocity v is given by $v = 4x^2 + 2$, where x is its displacement from the origin.

Which of these expressions represents the acceleration of the particle?

- A. $8x$
 - B. $32x^3 + 16x$
 - C. $8x^4 + 8x^2 + 2$
 - D. $16x^4 + 16x^2 + 4$
- 6 Let P and Q be the points with position vectors $\underline{p} = 3\underline{i} + 4\underline{j} - 2\underline{k}$ and $\underline{q} = -\underline{i} - 8\underline{j} + 4\underline{k}$ respectively.

What is the Cartesian equation of the sphere which has PQ as a diameter of the sphere?

- A. $(x - 1)^2 + (y + 2)^2 + (z - 1)^2 = 49$
- B. $(x + 2)^2 + (y + 6)^2 + (z - 3)^2 = 49$
- C. $(x - 1)^2 + (y + 2)^2 + (z - 1)^2 = 196$
- D. $(x + 2)^2 + (y + 6)^2 + (z - 3)^2 = 196$

- 7 Let ω be a non-real complex fifth root of unity.

What is the value of the expression $(1 - \omega)(1 - \omega^2)(1 - \omega^3)(1 - \omega^4)$?

- A. 0
- B. 1
- C. 4
- D. 5

- 8 The polynomial $P(z)$ has real coefficients. The complex number α is of the form $a + ib$, where a and b are both real, non-zero and distinct.

If $P(a)$, $P'(a)$, $P(b)$, $P'(b)$ and $P(\alpha)$ are all zero, what is the minimum degree of $P(z)$?

- A. 4
- B. 5
- C. 6
- D. 7

- 9 Which of the following is a primitive of $\int \sin^4 x \cos^5 x \, dx$?

- A. $\frac{1}{6} \cos^6 x + C$
- B. $\frac{1}{80} \sin^5 2x + C$
- C. $\frac{1}{5} \sin^5 x - \frac{2}{7} \sin^7 x + \frac{1}{9} \sin^9 x + C$
- D. $\frac{1}{6} \cos^6 x - \frac{1}{4} \cos^8 x + \frac{1}{10} \cos^{10} x + C$

- 10 A complex number z is defined such that $|z - i| = |z - \sqrt{3} + 2i|$.

What is the value of $\text{Arg}(z)$ when $|z|$ is a minimum?

- A. $-\frac{\pi}{3}$
- B. $-\frac{\pi}{6}$
- C. $\frac{\pi}{6}$
- D. $\frac{\pi}{3}$

Section II

90 marks

Attempt Questions 11–16

Allow about 2 hours and 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use the Question 11 Writing Booklet

- a) The location of the complex number z is shown on the diagram on page 1 of the Question 11 Writing Booklet. 2

On the diagram provided in the writing booklet, indicate the locations of z^2 and $-iz$.

- b) Consider the vector $\vec{v} = 3\vec{i} - 4\vec{j} + 12\vec{k}$. 1

Find a unit vector in the direction of $\vec{v}$.

- c) Let $z = 1 - i$.
- (i) Express z in modulus-argument form. 2
- (ii) Find z^{10} and express your answer in Cartesian form. 2

- d) A particle moves such that its position vector at time t is given by $\vec{r}(t) = (t^2)\vec{i} + (2t)\vec{j} + (\ln t)\vec{k}$ for $t > 0$. 2

Determine the time t at which the particle's velocity vector is perpendicular to its acceleration vector.

Question 11 continues on page 6

Question 11 (continued)

e) A polynomial $P(z) = z^3 + az^2 + bz + c$ has real coefficients a, b, c . **1**

If $z_1 = 2 + i$ and $z_2 = 3$ are roots of $P(z)$, what is the value of the coefficient b ?

f) Prove by contradiction that $\log_2 5$ is an irrational number. **2**

g) By using a suitable substitution, or otherwise, evaluate $\int_0^1 \sqrt{4 - x^2} dx$. **3**

End of Question 11

Question 12 (15 marks) Use the Question 12 Writing Booklet

a) Let a, b, c be positive real numbers.

(i) Show that $\frac{a+b}{2} \geq \sqrt{ab}$. 1

(ii) Prove that, if $a + b + c = 1$, then $(1 - a)(1 - b)(1 - c) \geq 8abc$. 2

b) A particle moves in a straight line with acceleration $a = 6x^2 - 4$, where x is the displacement from the origin. The particle starts from rest at $x = 1$. 3

Show that the particle is at rest at three different positions.

c) Find $\int x e^{2x} dx$. 2

d) (i) Find the values of A, B and C such that: 2

$$\frac{8}{x(x^2 + 4)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 4}$$

(ii) Hence, evaluate the definite integral: 2

$$\int_1^2 \frac{8}{x(x^2 + 4)} dx$$

e) The points A, B and C have position vectors $\underline{a} = \underline{i} + 2\underline{j} + 3\underline{k}$, $\underline{b} = 3\underline{i} - 4\underline{j} + 5\underline{k}$ and $\underline{c} = 2\underline{i} + \underline{j} + 6\underline{k}$ respectively. 3

Find the size of $\angle ABC$ to the nearest degree.

End of Question 12

Question 13 (15 marks) Use the Question 13 Writing Booklet

- a) Given that $|z - 1| = 2$ and $|w + 2| = 1$, find the maximum possible value of $|z - w|$. 2
Justify your answer.

- b) Let $\vec{u}$ and $\vec{v}$ be two non-zero vectors. 2

Prove that if the vectors $\vec{u} + \vec{v}$ and $\vec{u} - \vec{v}$ are perpendicular, then $|\vec{u}| = |\vec{v}|$.

- c) A line L has the vector equation $\vec{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$. 3

A sphere S has the equation $(x - 4)^2 + (y - 5)^2 + (z - 1)^2 = 9$.

Show that L and S do not have any points of intersection.

- d) By considering $e^{i(A+B)}$ and using Euler's formula, prove the trigonometric identity: 2

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

- e) Evaluate $\int_2^5 \frac{x-1}{\sqrt{5+4x-x^2}} dx$. 3

- f) Let z be a complex number such that $|z| = 1$. 3

Prove that $\frac{z-1}{z+1}$ is a purely imaginary number, given $z \neq -1$.

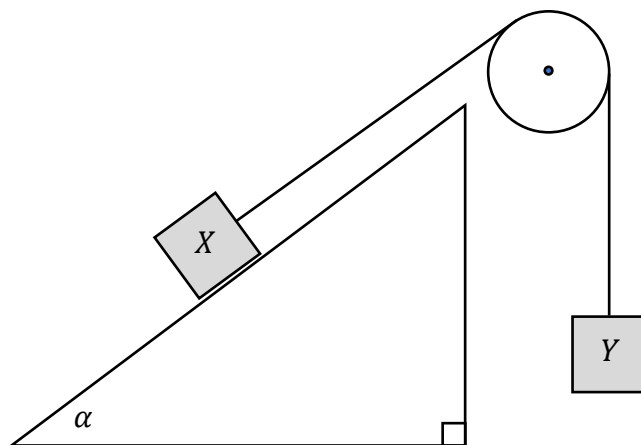
End of Question 13

Question 14 (15 marks) Use the Question 14 Writing Booklet

- a) (i) Sketch the region in the complex plane where both inequalities $|z - 2| \leq 2$ and $0 \leq \arg(z) \leq \frac{\pi}{4}$ are satisfied. 2
- (ii) Find the exact area of this region. 2
- b) Solve $z^2 - (2 + i)z + (3 + i) = 0$. 2
- c) An object, X , of mass 5 kg lies on a smooth plane inclined at an angle α to the horizontal, where $\sin \alpha = \frac{3}{5}$.

Object X is connected by a light inextensible string, which passes over a smooth pulley at the top of the plane, to a second object, Y , of mass 3 kg that hangs freely.

The system is released from rest.



- (i) Show that both objects are in equilibrium. 2
- (ii) Find the tension in the string. Use $g = 9.8 \text{ ms}^{-2}$. 1

Question 14 continues on page 10

Question 14 (continued)

- d) The points A , B and C in the complex plane represent the complex numbers z_1 , z_2 and z_3 respectively. **3**

If $\frac{z_2 - z_1}{z_3 - z_1} = i$, show that triangle ABC is a right-angled isosceles triangle.

- e) Prove by mathematical induction that, for all non-negative integers n , **3**

$$(x + a)^n = \sum_{r=0}^n \binom{n}{r} x^{n-r} a^r$$

End of Question 14

Question 15 (15 marks) Use the Question 15 Writing Booklet

a) Let z be a root of the equation $z^5 = -1$.

(i) Show that the roots of this equation are given by $z_k = e^{i\left(\frac{\pi}{5} + \frac{2k\pi}{5}\right)}$,
for $k = 0, 1, 2, 3, 4$. 2

(ii) Hence, prove that $\cos\left(\frac{\pi}{5}\right)\cos\left(\frac{3\pi}{5}\right) = -\frac{1}{4}$. 3

b) The polynomial $P(z) = z^4 - 4z^3 + 14z^2 - 36z + 45$ has a purely imaginary root. 3
Find all the roots of $P(z) = 0$.

c) Let a, b, c be the side lengths of a triangle. 3

Prove that the following inequality must hold:

$$\frac{a}{b+c-a} + \frac{b}{c+a-b} + \frac{c}{a+b-c} \geq 3$$

d) (i) Show that $\int_0^a f(x) dx = \int_0^a f(a-x) dx$. 2

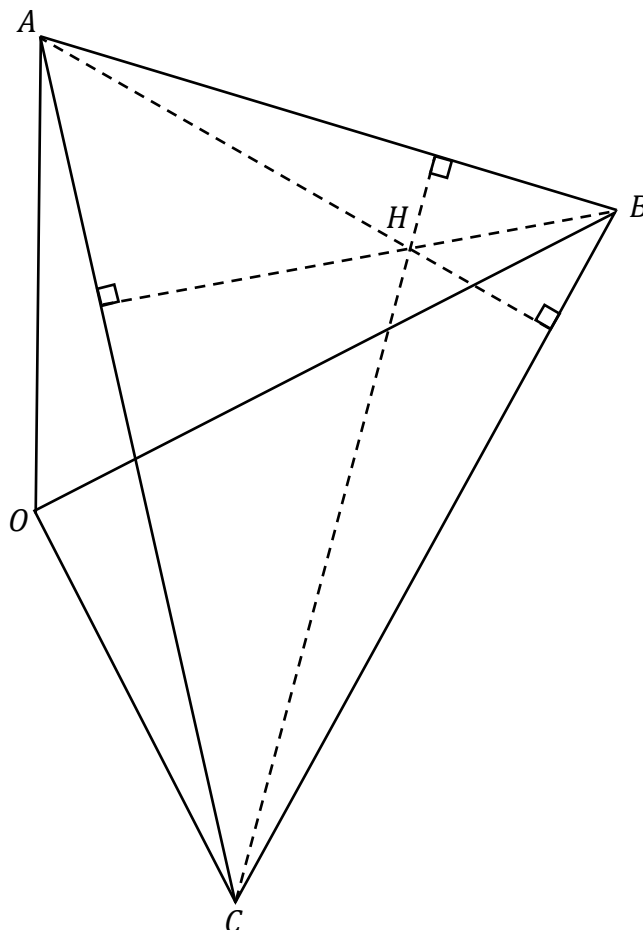
(iii) Hence evaluate $\int_0^{\frac{\pi}{2}} \frac{1}{1+\tan(x)} dx$. 2

End of Question 15

Question 16 (15 marks) Use the Question 16 Writing Booklet

- a) In the tetrahedron $OABC$, the edges OA , OB and OC are mutually perpendicular. 4
 Let $\overrightarrow{OA} = \underline{a}$, $\overrightarrow{OB} = \underline{b}$, $\overrightarrow{OC} = \underline{c}$.

The altitude of a triangle is the perpendicular line segment drawn from any vertex to the opposite side. Every triangle has three altitudes, which intersect at a single point called the orthocentre. Let H be the orthocentre of triangle ABC .



Similarly, the altitude of a pyramid is the perpendicular line segment drawn from any vertex to the opposite face. Consider the altitude of pyramid $OABC$, drawn from O and perpendicular to triangle ABC . Use vector methods to show that this altitude passes through H .

Question 16 continues on page 13

Question 16 (continued)

b) Find $\int \frac{1}{(x^2+1)^2} dx$. 3

c) Prove that, for any real numbers a and b , $a^2 + b^2 - ab - a - b + 1 \geq 0$. 2

d) Consider the integral $I_n = \int_0^1 (1-x^2)^n dx$ for integers $n \geq 0$.

(i) Show that for $n \geq 1$, the recurrence relation $I_n = \frac{2n}{2n+1} I_{n-1}$ holds. 3

(ii) Hence, show that $I_n = \frac{4^n (n!)^2}{(2n+1)!}$. 3

End of paper



SOLUTIONS

2026 HIGHER SCHOOL CERTIFICATE TRIAL EXAMINATION

Mathematics Extension 2

General instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Stick or write your Student Number at the top of this page and on each answer booklet

Total marks: 100

Section I – 10 marks (pages 2–4)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 5–13)

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

Section I

Multiple-choice answer key

1	D
2	A
3	C
4	B
5	B
6	A
7	D
8	C
9	C
10	A

- 1 Given $z_1 = 4e^{\frac{i\pi}{3}}$ and $z_2 = 2e^{\frac{i\pi}{6}}$, which of the following is the complex number $\frac{z_1}{z_2}$ expressed in Cartesian form?

- A. $2(1 + i\sqrt{3})$
- B. $2(\sqrt{3} + i)$
- C. $1 + i\sqrt{3}$
- D. $\sqrt{3} + i$

Sample solution:

$$\begin{aligned}\frac{z_1}{z_2} &= \frac{4}{2} e^{i(\frac{\pi}{3} - \frac{\pi}{6})} \\ &= 2e^{\frac{i\pi}{6}} \\ &= 2\left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right) \\ &= \sqrt{3} + i\end{aligned}$$

- 2 Consider the statement:

‘If a triangle is equilateral, then it is isosceles’.

Which of the following is the converse of that statement?

- A. If a triangle is isosceles, then it is equilateral.
- B. A triangle is equilateral if and only if it is isosceles.
- C. If a triangle is not equilateral, then it is not isosceles.
- D. If a triangle is not isosceles, then it is not equilateral.

Sample solution:

Converse of $P \Rightarrow Q$ is $Q \Rightarrow P$

Hence, converse is ‘if a triangle is isosceles, then it is equilateral’.

3 The three cube roots of a complex number w are z_1 , z_2 and z_3 .

If $z_1 = 2i$, what are the other two roots?

- A. $1 - i\sqrt{3}$ and $-1 - i\sqrt{3}$
- B. $-1 - i\sqrt{3}$ and $-1 + i\sqrt{3}$
- C. $\sqrt{3} - i$ and $-\sqrt{3} - i$
- D. $-\sqrt{3} - i$ and $-\sqrt{3} + i$

Sample solution:

$$2i = 2 \operatorname{cis} \frac{\pi}{2}$$

Other two roots will be $2 \operatorname{cis} \left(\frac{\pi}{2} \pm \frac{2\pi}{3} \right)$

$$2 \operatorname{cis} \left(\frac{\pi}{2} + \frac{2\pi}{3} \right) = 2 \operatorname{cis} \frac{7\pi}{6}$$

$$= 2 \left(-\frac{\sqrt{3}}{2} - \frac{1}{2}i \right)$$

$$= -\sqrt{3} - i$$

$$2 \operatorname{cis} \left(\frac{\pi}{2} - \frac{2\pi}{3} \right) = 2 \operatorname{cis} \left(-\frac{\pi}{6} \right)$$

$$= 2 \left(\frac{\sqrt{3}}{2} - \frac{1}{2}i \right)$$

$$= \sqrt{3} - i$$

4 Consider the statement:

$$\forall x \in \mathbb{Z}, \ln|x| > 0$$

Which of the following is the negation of this statement?

- A. $\exists x \in \mathbb{Z}$, such that $\ln|x| > 0$
- B. $\exists x \in \mathbb{Z}$, such that $\ln|x| \leq 0$
- C. $\forall x \in \mathbb{Z}$, $\ln|x| \leq 0$
- D. $\ln|x| > 0 \Rightarrow x \in \mathbb{Z}$

Sample solution:

Original statement: For all integers x , $\ln|x| > 0$

Negation of a 'for all' statement is a 'there exists' statement.

Negation: There exists an integer x such that $\ln|x| \leq 0$

- 5 A particle's velocity v is given by $v = 4x^2 + 2$, where x is its displacement from the origin.

Which of these expressions represents the acceleration of the particle?

- A. $8x$
- B. $32x^3 + 16x$
- C. $8x^4 + 8x^2 + 2$
- D. $16x^4 + 16x^2 + 4$

Sample solution:

$$\frac{dv}{dx} = 8x$$

$$\begin{aligned} a &= v \frac{dv}{dx} \\ &= (4x^2 + 2)(8x) \\ &= 32x^3 + 16x \end{aligned}$$

- 6 Let P and Q be the points with position vectors $\underline{p} = 3\hat{i} + 4\hat{j} - 2\hat{k}$ and $\underline{q} = -\hat{i} - 8\hat{j} + 4\hat{k}$ respectively.

What is the Cartesian equation of the sphere which has PQ as a diameter of the sphere?

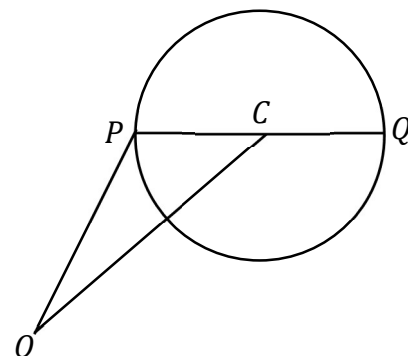
- A. $(x - 1)^2 + (y + 2)^2 + (z - 1)^2 = 49$
- B. $(x + 2)^2 + (y + 6)^2 + (z - 3)^2 = 49$
- C. $(x - 1)^2 + (y + 2)^2 + (z - 1)^2 = 196$
- D. $(x + 2)^2 + (y + 6)^2 + (z - 3)^2 = 196$

Sample solution:

$$\overrightarrow{PQ} = \begin{pmatrix} -4 \\ -12 \\ 6 \end{pmatrix}$$

$$\begin{aligned} |\overrightarrow{PQ}| &= \sqrt{16 + 144 + 36} \\ &= 14 \\ r &= 7 \end{aligned}$$

$$\begin{aligned} \text{Centre: } \overrightarrow{OC} &= \overrightarrow{OP} + \frac{1}{2}\overrightarrow{PQ} \\ &= \begin{pmatrix} 3 \\ 4 \\ -2 \end{pmatrix} + \begin{pmatrix} -2 \\ -6 \\ 3 \end{pmatrix} \\ &= \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} \end{aligned}$$



Hence equation is $(x - 1)^2 + (y + 2)^2 + (z - 1)^2 = 49$

7 Let ω be a non-real complex fifth root of unity.

What is the value of the expression $(1 - \omega)(1 - \omega^2)(1 - \omega^3)(1 - \omega^4)$?

- A. 0
- B. 1
- C. 4
- D. 5

Sample solution:

$P(z) = z^5 - 1$, roots are $1, \omega, \omega^2, \omega^3, \omega^4$

Consider two different factorisations of $P(z)$:

$$(z - 1)(z - \omega)(z - \omega^2)(z - \omega^3)(z - \omega^4) = (z - 1)(z^4 + z^3 + z^2 + z + 1)$$

Divide through by $(z - 1)$:

$$(z - \omega)(z - \omega^2)(z - \omega^3)(z - \omega^4) = z^4 + z^3 + z^2 + z + 1$$

Let $z = 1$:

$$\begin{aligned}(1 - \omega)(1 - \omega^2)(1 - \omega^3)(1 - \omega^4) &= 1 + 1 + 1 + 1 + 1 \\ &= 5\end{aligned}$$

8 The polynomial $P(z)$ has real coefficients. The complex number α is of the form $a + ib$, where a and b are both real, non-zero and distinct.

If $P(a), P'(a), P(b), P'(b)$ and $P(\alpha)$ are all zero, what is the minimum degree of $P(z)$?

- A. 4
- B. 5
- C. 6
- D. 7

Sample solution:

$P(a), P'(a)$ are zero $\Rightarrow a$ is a double root

$P(b), P'(b)$ are zero $\Rightarrow b$ is a double root

$P(\alpha) = 0 \Rightarrow P(\bar{\alpha}) = 0$, so both α and $\bar{\alpha}$ are roots

Hence $P(z)$ has degree at least 6

9 Which of the following is a primitive of $\int \sin^4 x \cos^5 x \, dx$?

- A. $\frac{1}{6} \cos^6 x + C$
- B. $\frac{1}{80} \sin^5 2x + C$
- C. $\frac{1}{5} \sin^5 x - \frac{2}{7} \sin^7 x + \frac{1}{9} \sin^9 x + C$
- D. $\frac{1}{6} \cos^6 x - \frac{1}{4} \cos^8 x + \frac{1}{10} \cos^{10} x + C$

Sample solution:

$$\begin{aligned} \int \sin^4 x \cos^5 x \, dx &= \int \cos x (1 - \sin^2 x)^2 \sin^4 x \, dx \\ &= \int \cos x (1 - 2 \sin^2 x + \sin^4 x) \sin^4 x \, dx \\ &= \int \cos x (\sin^4 x - 2 \sin^6 x + \sin^8 x) \, dx \\ &= \frac{1}{5} \sin^5 x - \frac{2}{7} \sin^7 x + \frac{1}{9} \sin^9 x + c \end{aligned}$$

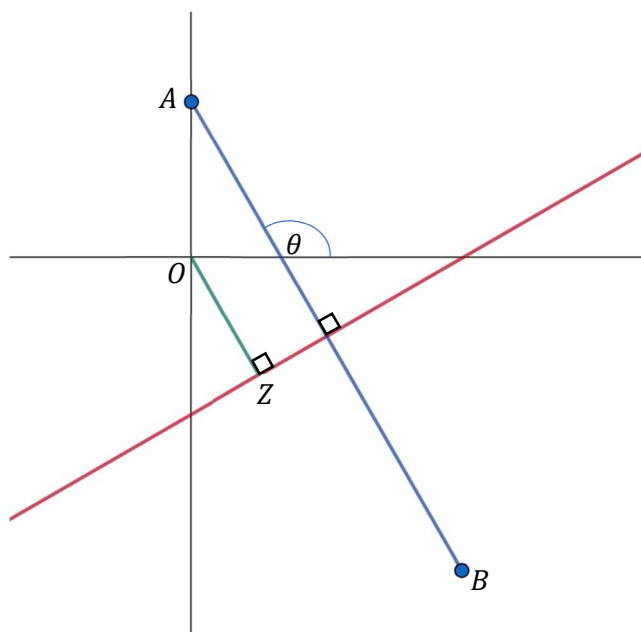
10 A complex number z is defined such that $|z - i| = |z - \sqrt{3} + 2i|$.

What is the value of $\text{Arg}(z)$ when $|z|$ is a minimum?

- A. $-\frac{\pi}{3}$
- B. $-\frac{\pi}{6}$
- C. $\frac{\pi}{6}$
- D. $\frac{\pi}{3}$

Sample solution:

z lies on the perpendicular bisector of the interval joining $A(0, 1)$ and $B(\sqrt{3}, -2)$.



When $|z|$ is a minimum, the interval OZ is perpendicular to the perpendicular bisector of the interval joining $A(0, 1)$ and $B(\sqrt{3}, -2)$. That is, OZ is parallel to AB .

$$m_{OZ} = m_{AB} = -\sqrt{3}$$

$$\tan \theta = -\sqrt{3}$$

$$\theta = \frac{2\pi}{3}$$

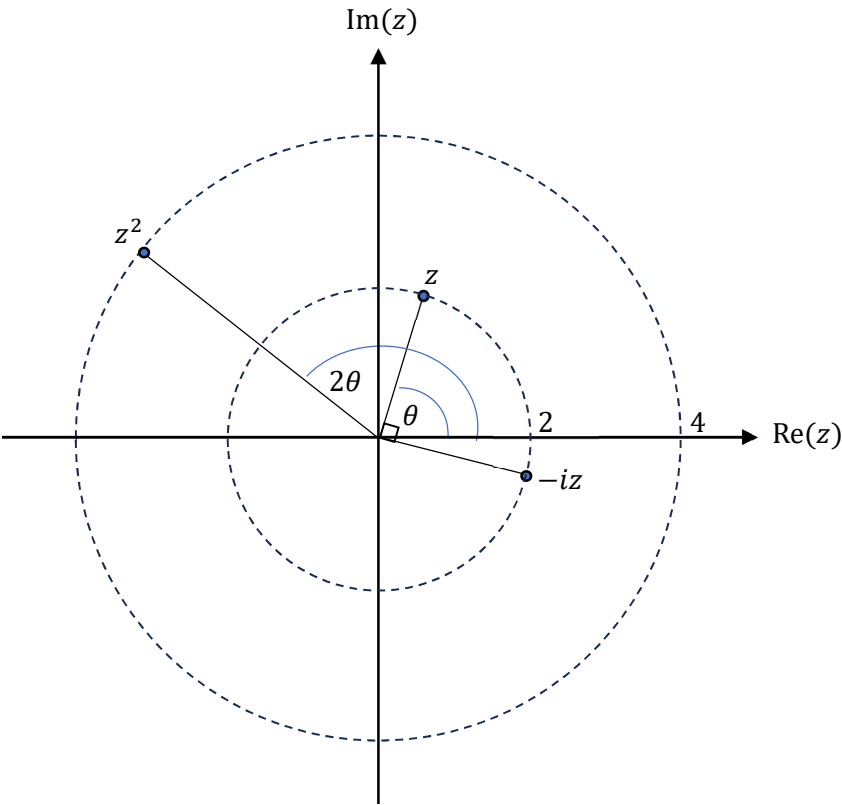
$$\text{Arg}(z) = -\frac{\pi}{3}$$

Question 11 (15 marks)

- a) The location of the complex number z is shown on the diagram on page 1 of the Question 11 Writing Booklet. 2

On the diagram provided in the writing booklet, indicate the locations of z^2 and $-iz$.

Sample solution:



Marks	Criteria
2	Both points correct, with necessary details labelled
1	One point correct with details labelled OR both points in correct vicinity, but necessary details not labelled

b) Consider the vector $\underline{v} = 3\underline{i} - 4\underline{j} + 12\underline{k}$.

1

Find a unit vector in the direction of $\underline{v}$.

Sample solution:

$$|\underline{v}| = \sqrt{3^2 + (-4)^2 + 12^2}$$

$$= 13$$

$$\hat{v} = \frac{1}{13}(3\underline{i} - 4\underline{j} + 12\underline{k})$$

Marks	Criteria
1	Correct answer

c) Let $z = 1 - i$.

(i) Express z in modulus-argument form.

2

Sample solution:

$$r = \sqrt{1^2 + 1^2}$$

$$= \sqrt{2}$$

$$\tan \theta = -1$$

$$\theta = -\frac{\pi}{4} \left(\text{or } \frac{7\pi}{4} \right)$$

$$z = \sqrt{2} \left(\cos \left(-\frac{\pi}{4} \right) + i \sin \left(-\frac{\pi}{4} \right) \right) \quad \text{or} \quad \sqrt{2} \left(\cos \left(\frac{7\pi}{4} \right) + i \sin \left(\frac{7\pi}{4} \right) \right)$$

Marks	Criteria
2	Provides correct solution (principal argument form not necessary)
1	Correct modulus or argument

(ii) Find z^{10} and express your answer in Cartesian form.

2

Sample solution:

$$\begin{aligned} z^{10} &= (\sqrt{2})^{10} \left(\cos \left(10 \times -\frac{\pi}{4} \right) + i \sin \left(10 \times -\frac{\pi}{4} \right) \right) \\ &= 32 \left(\cos \left(-\frac{5\pi}{2} \right) + i \sin \left(-\frac{5\pi}{2} \right) \right) \\ &= 32(0 - i) \\ &= -32i \end{aligned}$$

Marks	Criteria
2	Provides correct solution
1	Correctly finds z^{10} in modulus-argument form OR finds correct modulus or argument and then correctly converts their z^{10} into Cartesian form

- d) A particle moves such that its position vector at time t is given by

2

$$\vec{r}(t) = (t^2)\vec{i} + (2t)\vec{j} + (\ln t)\vec{k} \text{ for } t > 0.$$

Determine the time t at which the particle's velocity vector is perpendicular to its acceleration vector.

Sample solution:

$$\vec{v}(t) = (2t)\vec{i} + (2)\vec{j} + \left(\frac{1}{t}\right)\vec{k}$$

$$\vec{a}(t) = (2)\vec{i} + (0)\vec{j} - \left(\frac{1}{t^2}\right)\vec{k}$$

Perpendicular when $\vec{v} \cdot \vec{a} = 0$

$$2t(2) + 2(0) + \frac{1}{t}\left(-\frac{1}{t^2}\right) = 0$$

$$4t - \frac{1}{t^3} = 0$$

$$4t^4 - 1 = 0$$

$$4t^4 = 1$$

$$t^4 = \frac{1}{4}$$

$$t = \pm \frac{1}{\sqrt{2}}$$

But $t \geq 0$, so $t = \frac{1}{\sqrt{2}}$

Marks	Criteria
2	Provides correct solution (accept $t = \pm \frac{1}{\sqrt{2}}$)
1	Finds correct acceleration vector and states $\vec{v} \cdot \vec{a} = 0$, or equivalent merit

- e) A polynomial $P(z) = z^3 + az^2 + bz + c$ has real coefficients a, b, c . 1

If $z_1 = 2 + i$ and $z_2 = 3$ are roots of $P(z)$, what is the value of the coefficient b ?

Sample solution:

Real coefficients $\Rightarrow 2 - i$ is also a root

Sum of roots $2 + 2 - i + 3 = b$

$$\begin{aligned} b &= (2 + i)(2 - i) + 3(2 + i) + 3(2 - i) \\ &= 4 + 1 + 6 + 3i + 6 - 3i \\ &= 17 \end{aligned}$$

Marks	Criteria
1	Correct answer

- f) Prove by contradiction that $\log_2 5$ is an irrational number. 2

Sample solution:

Assume $\log_2 5$ is a rational number

i.e. $\log_2 5 = \frac{p}{q}$ where $p, q \in \mathbb{Z}$, $q \neq 0$ and p, q are coprime.

$$\log_2 5 = \frac{p}{q}$$

$$2^{\frac{p}{q}} = 5$$

$$2^p = 5^q$$

But since 2 and 5 are both prime numbers, this cannot be true.

Hence the assumption is false, so $\log_2 5$ is irrational.

Marks	Criteria
2	Provides complete and logical proof
1	Significant progress, but minor errors made or minor details missing

g) By using a suitable substitution, or otherwise, evaluate $\int_0^1 \sqrt{4-x^2} dx$.

3

Sample solution:

Let $x = 2 \sin \theta$

$$\frac{dx}{d\theta} = 2 \cos \theta$$

$$dx = 2 \cos \theta d\theta$$

When $x = 0$, $\theta = 0$

When $x = 1$, $\theta = \frac{\pi}{6}$

$$\begin{aligned} \int_0^1 \sqrt{4-x^2} dx &= \int_0^{\frac{\pi}{6}} \sqrt{4-4\sin^2 \theta} \cdot 2 \cos \theta d\theta \\ &= \int_0^{\frac{\pi}{6}} \sqrt{4\cos^2 \theta} \cdot 2 \cos \theta d\theta \\ &= \int_0^{\frac{\pi}{6}} 4 \cos^2 \theta d\theta \\ &= 2 \int_0^{\frac{\pi}{6}} 1 + \cos 2\theta d\theta \\ &= 2 \left[\theta + \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{6}} \\ &= 2 \left(\frac{\pi}{6} + \frac{1}{2} \sin \frac{\pi}{3} - \left(0 + \frac{1}{2} \sin 0 \right) \right) \\ &= 2 \left(\frac{\pi}{6} + \frac{\sqrt{3}}{4} - 0 \right) \\ &= \frac{\pi}{3} + \frac{\sqrt{3}}{2} \end{aligned}$$

Marks	Criteria
3	Provides correct solution
2	Obtains correct primitive OR correct definite integral in terms of θ , or equivalent merit
1	Attempts to use a suitable substitution or equivalent merit

Question 12 (15 marks)

a) Let a, b, c be positive real numbers.

(i) Show that $\frac{a+b}{2} \geq \sqrt{ab}$.

1

Sample solution:

$$(\sqrt{a} - \sqrt{b})^2 \geq 0$$

$$a - 2\sqrt{ab} + b \geq 0$$

$$a + b \geq 2\sqrt{ab}$$

$$\frac{a+b}{2} \geq \sqrt{ab}$$

Marks	Criteria
1	Provides correct proof

(ii) Prove that, if $a + b + c = 1$, then $(1 - a)(1 - b)(1 - c) \geq 8abc$.

2

Sample solution:

$$1 - a = b + c$$

$$1 - b = a + c$$

$$1 - c = a + b$$

$$(1 - a)(1 - b)(1 - c) = (b + c)(a + c)(a + b)$$

$$\geq 2\sqrt{bc} \times 2\sqrt{ac} \times 2\sqrt{ab} \quad (\text{using part i})$$

$$\geq 8\sqrt{a^2b^2c^2}$$

$$\geq 8abc$$

Marks	Criteria
2	Provides correct proof
1	Attempts to use appropriate expressions for $(1 - a)$, $(1 - b)$ and $(1 - c)$ OR successfully incorporates result from part (i), or equivalent merit

- b) A particle moves in a straight line with acceleration $a = 6x^2 - 4$, where x is the displacement from the origin. The particle starts from rest at $x = 1$. **3**

Show that the particle is at rest at three different positions.

Sample solution:

$$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 6x^2 - 4$$

$$\frac{1}{2} v^2 = \int 6x^2 - 4 \, dx$$

$$\frac{1}{2} v^2 = 2x^3 - 4x + c$$

When $x = 1$, $v = 0$

$$0 = 2(1) - 4(1) + c$$

$$c = 2$$

$$\frac{1}{2} v^2 = 2x^3 - 4x + 2$$

$$v^2 = 4x^3 - 8x + 4$$

At rest when $v = 0$

$$0 = 4x^3 - 8x + 4$$

$$0 = x^3 - 2x + 1$$

At rest when $x = 1$, so $(x - 1)$ is a factor

$$\begin{array}{r} x^2 + x - 1 \\ x-1 \overline{) x^3 + 0x^2 - 2x + 1} \\ \underline{x^3 - x^2} \\ x^2 - 2x + 1 \\ \underline{x^2 - x} \\ -x + 1 \\ \underline{-x + 1} \\ 0 \end{array}$$

$$0 = (x - 1)(x^2 + x - 1)$$

$$x = 1, \frac{-1 \pm \sqrt{5}}{2}$$

i.e. the particle is at rest in 3 different positions

Marks	Criteria
3	Provides correct solution
2	Forms a cubic equation and attempts to use polynomial division or sums/products of roots to find the factors, or equivalent merit
1	Finds a correct expression for v^2 , including finding constant of integration

c) Find $\int x e^{2x} dx$.

2

Sample solution:

$$\begin{aligned} \text{Let } u &= x & v' &= e^{2x} \\ u' &= 1 & v &= \frac{1}{2} e^{2x} \end{aligned}$$

$$\begin{aligned} \int x e^{2x} dx &= \left[\frac{1}{2} x e^{2x} \right] - \frac{1}{2} \int e^{2x} dx \\ &= \frac{1}{2} x e^{2x} - \frac{1}{2} \left(\frac{1}{2} e^{2x} + c \right) \\ &= \frac{1}{2} x e^{2x} - \frac{1}{4} e^{2x} + C \end{aligned}$$

Marks	Criteria
2	Provides correct solution (no penalty for omitting +c)
1	Attempts to use integration by parts

- d) (i) Find the values of A , B and C such that:

2

$$\frac{8}{x(x^2 + 4)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 4}$$

Sample solution:

$$8 = A(x^2 + 4) + x(Bx + C)$$

$$8 = Ax^2 + 4A + Bx^2 + Cx$$

$$4A = 8 \Rightarrow A = 2$$

$$A + B = 0 \Rightarrow B = -2$$

$$C = 0$$

Marks	Criteria
2	Provides correct solution
1	Evaluates one of the three coefficients

- (ii) Hence, evaluate the definite integral:

2

$$\int_1^2 \frac{8}{x(x^2 + 4)} dx$$

Sample solution:

$$\begin{aligned} \int_1^2 \frac{8}{x(x^2 + 4)} dx &= \int_1^2 \frac{2}{x} - \frac{2x}{x^2 + 4} dx \\ &= [2 \ln x - \ln(x^2 + 4)]_1^2 \\ &= 2 \ln 2 - \ln 8 - (2 \ln 1 - \ln 5) \\ &= \ln 4 - \ln 8 + \ln 5 \\ &= \ln \frac{5}{2} \end{aligned}$$

Marks	Criteria
2	Provides correct solution (accept answer as a sum/difference of logs)
1	Finds a correct primitive, or equivalent merit

- e) The points A , B and C have position vectors $\vec{a} = \vec{i} + 2\vec{j} + 3\vec{k}$, $\vec{b} = 3\vec{i} - 4\vec{j} + 5\vec{k}$ and $\vec{c} = 2\vec{i} + \vec{j} + 6\vec{k}$ respectively.

3

Find the size of $\angle ABC$ to the nearest degree.

Sample solution:

$$\vec{BA} = -2\hat{i} + 6\hat{j} - 2\hat{k}$$

$$\vec{BC} = -\hat{i} + 5\hat{j} + \hat{k}$$

$$|\vec{BA}| = \sqrt{4 + 36 + 4} = \sqrt{44}$$

$$|\vec{BC}| = \sqrt{1 + 25 + 1} = \sqrt{27}$$

$$\begin{aligned}\cos(\angle ABC) &= \frac{-2(-1) + 6(5) - 2(1)}{\sqrt{44} \times \sqrt{27}} \\ &= \frac{30}{\sqrt{1188}}\end{aligned}$$

$$\angle ABC = 29^\circ$$

Marks	Criteria
3	Provides correct solution
2	Finds correct expressions for $\vec{BA}$, $\vec{BC}$, $ \vec{BA} $ and $ \vec{BC} $, or equivalent merit
1	Finds correct expressions for $\vec{BA}$ and $\vec{BC}$ OR finds correct $ \vec{BA} $ and $ \vec{BC} $ OR finds correct $\vec{BC}$ and $ \vec{BC} $

Question 13 (15 marks)

- a) Given that $|z - 1| = 2$ and $|w + 2| = 1$, find the maximum possible value of $|z - w|$. **2**
Justify your answer.

Sample solution 1:

$$|z - w| = |(z - 1) - (w + 2) + 3|$$

Triangle inequality: $|x \pm y| \leq |x| + |y|$

Hence:

$$|z - w| \leq |(z - 1) + 3| + |(w + 2)|$$

$$|z - w| \leq |(z - 1)| + |3| + |(w + 2)|$$

$$|z - w| \leq 2 + 3 + 1$$

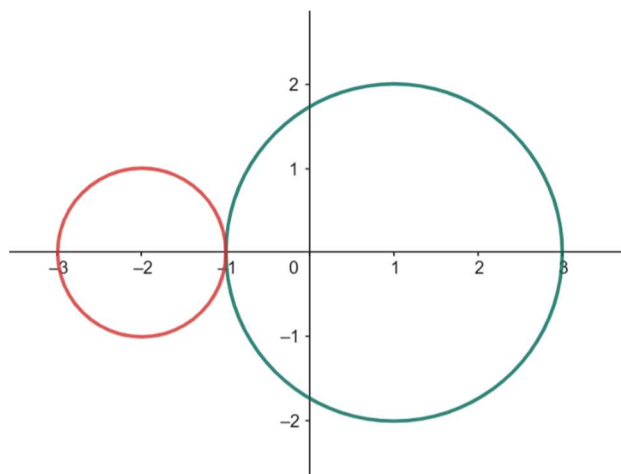
$$|z - w| \leq 6$$

Sample solution 2:

Maximum value of $|z - w|$ occurs when z and w lie in a straight line.

From the diagram, this happens when $z = 3$ and $w = -3$.

Hence $|z - w| \leq |3 - (-3)| = 6$.



Marks	Criteria
2	Provides correct solution with suitable justification
1	Provides a correct triangle inequality statement or draws both circles correctly on the same plane or equivalent measure.

b) Let $\underline{u}$ and $\underline{v}$ be two non-zero vectors.

2

Prove that if the vectors $\underline{u} + \underline{v}$ and $\underline{u} - \underline{v}$ are perpendicular, then $|\underline{u}| = |\underline{v}|$.

Sample solution:

$$(\underline{u} + \underline{v}) \cdot (\underline{u} - \underline{v}) = 0$$

$$\underline{u} \cdot \underline{u} - \underline{u} \cdot \underline{v} + \underline{v} \cdot \underline{u} - \underline{v} \cdot \underline{v} = 0$$

$$|\underline{u}|^2 - |\underline{v}|^2 = 0$$

$$|\underline{u}|^2 = |\underline{v}|^2$$

$$|\underline{u}| = |\underline{v}| \quad \text{since } |\underline{u}|, |\underline{v}| \geq 0$$

Marks	Criteria
2	Provides correct proof
1	Provides correct expansion of $(\underline{u} + \underline{v}) \cdot (\underline{u} - \underline{v})$ and equates this to zero

c) A line L has the vector equation $\vec{r} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}$.

3

A sphere S has the equation $(x - 4)^2 + (y - 5)^2 + (z - 1)^2 = 9$.

Show that L and S do not have any points of intersection.

Sample solution:

$L: x = 1 + 2\lambda$

$y = 2 - \lambda$

$z = 3 + 2\lambda$

For points of intersection, sub into equations of S .

$$(1 + 2\lambda - 4)^2 + (2 - \lambda - 5)^2 + (3 + 2\lambda - 1)^2 = 9$$

$$(2\lambda - 3)^2 + (-3 - \lambda)^2 + (2 + 2\lambda)^2 = 9$$

$$4\lambda^2 - 12\lambda + 9 + 9 + 6\lambda + \lambda^2 + 4 + 8\lambda + 4\lambda^2 = 9$$

$$9\lambda^2 + 2\lambda + 13 = 0$$

$$\Delta = b^2 - 4ac$$

$$= 2^2 - 4(9)(13)$$

$$= -464$$

$$< 0 \quad \therefore \text{no solutions for } \lambda$$

Hence L and S do not intersect.

Marks	Criteria
3	Provides correct solution
2	Forms a simplified quadratic equation in λ
1	Forms parametric equations for the components of L and substitutes these into the equation for S , or equivalent merit

- d) By considering $e^{i(A+B)}$ and using Euler's formula, prove the trigonometric identity: 2

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

Sample solution:

$$e^{i(A+B)} = e^{iA} \times e^{iB}$$

Using Euler's formula:

$$\begin{aligned} \cos(A + B) + i \sin(A + B) &= (\cos A + i \sin A)(\cos B + i \sin B) \\ &= \cos A \cos B + i \cos A \sin B + i \sin A \cos B - \sin A \sin B \\ &= \cos A \cos B - \sin A \sin B + i(\cos A \sin B + \sin A \cos B) \end{aligned}$$

Equating real parts:

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

Marks	Criteria
2	Provides correct solution
1	Provides two different expressions for $e^{i(A+B)}$ in modulus-argument form, or equivalent merit Completes a valid proof which equates $e^{i(A+B)}$ to $\text{cis}(A + B)$

e) Evaluate $\int_2^5 \frac{x-1}{\sqrt{5+4x-x^2}} dx$.

3

Sample solution:

$$\begin{aligned}
 \int_2^5 \frac{x-1}{\sqrt{5+4x-x^2}} dx &= -\frac{1}{2} \int_2^5 \frac{4-2x-2}{\sqrt{5+4x-x^2}} dx \\
 &= -\frac{1}{2} \int_2^5 (4-2x)(5+4x-x^2)^{-\frac{1}{2}} - \frac{2}{\sqrt{9-(x^2-4x+4)}} dx \\
 &= -\frac{1}{2} \left[2(5+4x-x^2)^{\frac{1}{2}} - 2 \sin^{-1} \left(\frac{x-2}{3} \right) \right]_2^5 \\
 &= -\frac{1}{2} \left(2\sqrt{0} - 2 \sin^{-1} 1 - \left(2(\sqrt{9} - 2 \sin^{-1} 0) \right) \right) \\
 &= -\frac{1}{2} \left(-2 \left(\frac{\pi}{2} \right) - 6 \right) \\
 &= \frac{\pi}{2} + 3
 \end{aligned}$$

Marks	Criteria
3	Provides correct solution
2	Splits the integral into the sum of two integrals and correctly integrates one, or equivalent merit
1	Completes the square for $5+4x-x^2$ OR writes $x-1 = \frac{1}{2}(4-2x-2)$, or equivalent merit

f) Let z be a complex number such that $|z| = 1$.

3

Prove that $\frac{z-1}{z+1}$ is a purely imaginary number, given $z \neq -1$.

Sample solution:

$$\begin{aligned}\frac{z-1}{z+1} &= \frac{x-1+iy}{x+1+iy} \times \frac{x+1-iy}{x+1-iy} \\ &= \frac{x-(1-iy)}{(x+1)+iy} \times \frac{x+(1-iy)}{(x+1)-iy} \\ &= \frac{x^2-(1-iy)^2}{(x+1)^2-(iy)^2} \\ &= \frac{x^2-(1-2iy-y^2)}{(x+1)^2+y^2} \\ &= \frac{x^2+y^2-1+2iy}{(x+1)^2+y^2}\end{aligned}$$

$$\operatorname{Re}\left(\frac{z-1}{z+1}\right) = \frac{x^2+y^2-1}{(x+1)^2+y^2}$$

$$\begin{aligned}\text{But } |z| &= \sqrt{x^2+y^2} = 1 \\ x^2+y^2 &= 1\end{aligned}$$

$$\therefore \operatorname{Re}\left(\frac{z-1}{z+1}\right) = \frac{1-1}{(x+1)^2+y^2} = 0$$

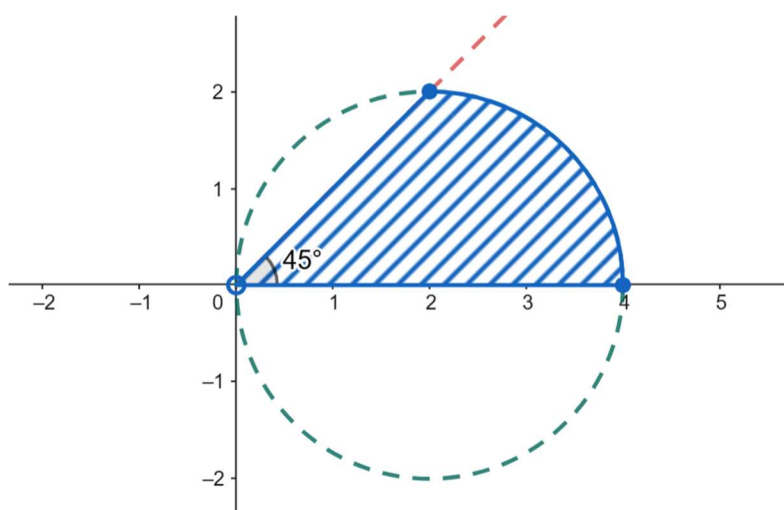
i.e. $\frac{z-1}{z+1}$ is purely imaginary

Marks	Criteria
3	Provides correct solution
2	Makes significant progress towards a correct solution
1	Expresses $\frac{z-1}{z+1}$ in terms of x and y and multiplies top and bottom by the conjugate of the denominator

Question 14 (15 marks)

- a) (i) Sketch the region in the complex plane where both inequalities $|z - 2| \leq 2$ and $0 \leq \arg(z) \leq \frac{\pi}{4}$ are satisfied. 2

Sample solution:



Marks	Criteria
2	Provides correct solution (no penalty if entire circle and ray are solid lines)
1	Correctly sketches one of the required regions, or equivalent merit

- (ii) Find the exact area of this region. 2

Sample solution:

$$\begin{aligned} \text{Area} &= \frac{1}{4} \times \pi \times 2^2 + \frac{1}{2} \times 2 \times 2 \\ &= (\pi + 2) u^2 \end{aligned}$$

Marks	Criteria
2	Provides correct solution (note that a student cannot gain full marks if their diagram in (i) did not result in a composite area)
1	Finds a correct expression for the area of the quadrant or the area of the triangle

- b) Solve $z^2 - (2 + i)z + (3 + i) = 0$. 2

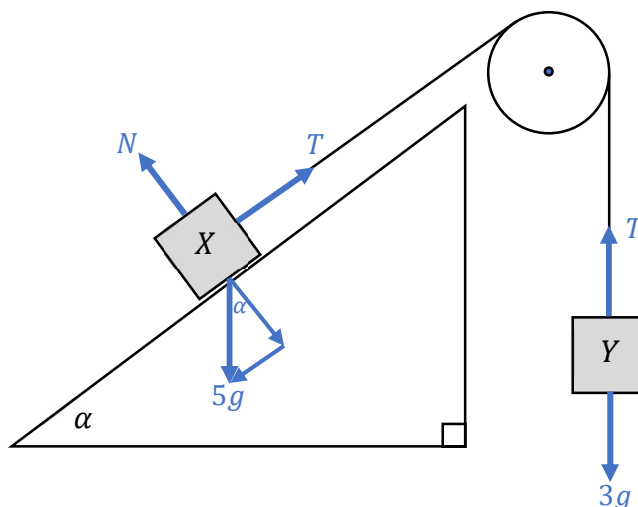
Sample solution:

$$z = \frac{2 + i \pm \sqrt{(2 + i)^2 - 4(3 + i)}}{2}$$

$$\begin{aligned}
&= \frac{2 + i \pm \sqrt{4 + 4i - 1 - 12 - 4i}}{2} \\
&= \frac{2 + i \pm \sqrt{-9}}{2} \\
&= \frac{2 + i \pm 3i}{2} \\
&= \frac{2 + 4i}{2}, \frac{2 - 2i}{2} \\
&= 1 + 2i, 1 - i
\end{aligned}$$

Marks	Criteria
2	Provides correct solution
1	Correctly substitutes into the quadratic formula and expands the discriminant, or equivalent merit

- c) An object, X , of mass 5 kg lies on a smooth plane inclined at an angle α to the horizontal, where $\sin \alpha = \frac{3}{5}$. Object X is connected by a light inextensible string, which passes over a smooth pulley at the top of the plane, to a second object, Y , of mass 3 kg that hangs freely. The system is released from rest.



- (i) Show that both objects are in equilibrium.

2

Sample solution:

Consider the forces in the direction of motion for each object. Since object X isn't falling through the plane, the forces on X perpendicular to the plane will cancel each other out.

Object X :

$$F = 5g \sin \alpha - T$$

$$5a = 5g \times \frac{3}{5} - T$$

$$5a = 3g - T$$

Object Y :

$$F = 3g - T$$

$$3a = 3g - T$$

Since acceleration for each object is equal:

$$5a = 3a$$

$$a = 0$$

Since the acceleration on each object is zero, the system is in equilibrium.

Marks	Criteria
2	Provides a correct and thorough solution
1	Decomposes the forces on object X into components parallel and perpendicular to the plane OR finds an expression for the resultant force on object Y OR completes solution but leaves out both tension forces, or equivalent merit

- (ii) Find the tension in the string. Use $g = 9.8 \text{ ms}^{-2}$.

1

Sample solution:

$$3g - T = 0$$

$$\begin{aligned}
 T &= 3g \\
 &= 3 \times 9.8 \\
 &= 29.4 \text{ N}
 \end{aligned}$$

Marks	Criteria
1	Provides correct answer

- d) The points A , B and C in the complex plane represent the complex numbers z_1 , z_2 and z_3 respectively. 3

If $\frac{z_2 - z_1}{z_3 - z_1} = i$, show that triangle ABC is a right-angled isosceles triangle.

Sample solution:

$$z_2 - z_1 = \overrightarrow{AB}$$

$$z_3 - z_1 = \overrightarrow{AC}$$

$$\left| \frac{z_2 - z_1}{z_3 - z_1} \right| = |i|$$

$$\frac{|z_2 - z_1|}{|z_3 - z_1|} = 1$$

$$|z_2 - z_1| = |z_3 - z_1|$$

$$|\overrightarrow{AB}| = |\overrightarrow{AC}|$$

Hence $\triangle ABC$ is isosceles

$$\arg\left(\frac{z_2 - z_1}{z_3 - z_1}\right) = \arg(i)$$

$$\arg(z_2 - z_1) - \arg(z_3 - z_1) = \frac{\pi}{2}$$

$$\arg(\overrightarrow{AB}) - \arg(\overrightarrow{AC}) = \frac{\pi}{2}$$

$$\angle BAC = \frac{\pi}{2}$$

Hence $\triangle ABC$ is right-angled at A

Marks	Criteria
3	Provides correct solution
2	Correctly shows that $\triangle ABC$ is isosceles OR that $\triangle ABC$ is right-angled at A
1	Correctly expresses $z_2 - z_1$ and $z_3 - z_1$ in terms of A , B and C

e) Prove by mathematical induction that, for all non-negative integers n ,

3

$$(x + a)^n = \sum_{r=0}^n \binom{n}{r} x^{n-r} a^r$$

Sample solution:

* Prove true for $n = 0$

$$LHS = (x + a)^0$$

$$= 1$$

$$RHS = \binom{0}{0} x^0 a^0$$

$$= 1$$

$\therefore$ true for $n = 0$

* Assume true for $n = k$

i.e. assume:

$$(x + a)^k = \binom{k}{0} x^k + \binom{k}{1} x^{k-1} a + \binom{k}{2} x^{k-2} a^2 + \dots + \binom{k}{k} a^k$$

* Prove true for $n = k + 1$, if true for $n = k$

i.e. prove:

$$(x + a)^{k+1} = \binom{k+1}{0} x^{k+1} + \binom{k+1}{1} x^k a + \binom{k+1}{2} x^{k-1} a^2 + \dots + \binom{k+1}{k+1} a^{k+1}$$

$$LHS = (x + a)(x + a)^k$$

$$= (x + a) \left(\binom{k}{0} x^k + \binom{k}{1} x^{k-1} a + \binom{k}{2} x^{k-2} a^2 + \dots + \binom{k}{k} a^k \right) \text{ by assumption}$$

$$= \left(\binom{k}{0} x^{k+1} + \binom{k}{1} x^k a + \dots + \binom{k}{k} x a^k \right) + \left(\binom{k}{0} x^k a + \binom{k}{1} x^{k-1} a^2 + \dots + \binom{k}{k} a^{k+1} \right) \quad **$$

$$= \binom{k}{0} x^{k+1} + \left(\binom{k}{0} + \binom{k}{1} \right) x^k a + \left(\binom{k}{1} + \binom{k}{2} \right) x^{k-1} a^2 + \dots + \left(\binom{k}{k-1} + \binom{k}{k} \right) x a^k + \binom{k}{k} a^{k+1}$$

But:

$$\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k}, \quad \binom{n}{0} = \binom{n+1}{0} = 1, \quad \binom{n}{n} = \binom{n+1}{n+1} = 1$$

Hence:

$$LHS = \binom{k+1}{0} x^{k+1} + \binom{k+1}{1} x^k a + \binom{k+2}{2} x^{k-1} a^2 + \dots + \binom{k+1}{k} x a^k + \binom{k+1}{k+1} a^{k+1}$$

Hence true for $n = k + 1$ if true for $n = k$.

Since true for $n = 0$, $\therefore$ true for all $n \geq 0$ by mathematical induction.

Marks	Criteria
3	Provides correct proof
2	Establishes the inductive step, or equivalent merit
1	Establishes the base case OR progresses in inductive step up to **, or equivalent merit

Question 15 (15 marks)

a) Let z be a root of the equation $z^5 = -1$.

- (i) Show that the roots of this equation are given by $z_k = e^{i\left(\frac{\pi}{5} + \frac{2k\pi}{5}\right)}$,
for $k = 0, 1, 2, 3, 4$.

2

Sample solution:

$$z^5 = -1$$

$$r^5 \operatorname{cis} 5\theta = \operatorname{cis} \pi$$

$$r^5 = 1$$

$$5\theta = \pi + 2k\pi, \quad k = 0, 1, 2, 3, 4$$

$$r = 1$$

$$\theta = \frac{\pi + 2k\pi}{5}$$

$$z_k = \operatorname{cis}\left(\frac{\pi}{5} + \frac{2k\pi}{5}\right)$$

$$z_k = e^{i\left(\frac{\pi}{5} + \frac{2k\pi}{5}\right)}$$

Marks	Criteria
2	Provides correct solution
1	Correctly shows $r = 1$ OR correctly shows $\theta = \frac{\pi + 2k\pi}{5}$

- (ii) Hence, prove that $\cos\left(\frac{\pi}{5}\right)\cos\left(\frac{3\pi}{5}\right) = -\frac{1}{4}$.

3

Sample solution:

$$z_0 = e^{\frac{i\pi}{5}}$$

$$z_1 = e^{\frac{3i\pi}{5}}$$

$$z_2 = e^{i\pi} = -1$$

$$z_3 = e^{\frac{7i\pi}{5}} = e^{-\frac{3i\pi}{5}}$$

$$z_4 = e^{\frac{9i\pi}{5}} = e^{-\frac{i\pi}{5}}$$

Consider one factorisation:

$$z^5 + 1 = 0$$

$$(z + 1) \left(z - e^{\frac{i\pi}{5}} \right) \left(z - e^{-\frac{i\pi}{5}} \right) \left(z - e^{\frac{3i\pi}{5}} \right) \left(z - e^{-\frac{3i\pi}{5}} \right) = 0$$

$$(z + 1) \left(z^2 - 2z \cos \frac{\pi}{5} + 1 \right) \left(z^2 - 2z \cos \frac{3\pi}{5} + 1 \right) = 0$$

Consider another factorisation:

$$z^5 + 1 = 0$$

$$(z + 1)(z^4 - z^3 + z^2 - z + 1) = 0$$

Hence:

$$z^4 - z^3 + z^2 - z + 1 = \left(z^2 - 2z \cos \frac{\pi}{5} + 1 \right) \left(z^2 - 2z \cos \frac{3\pi}{5} + 1 \right)$$

Consider the term in z^2 on each side:

$$z^2 = z^2 + 4z^2 \cos \frac{\pi}{5} \cos \frac{3\pi}{5} + z^2$$

$$1 = 2 + 4 \cos \frac{\pi}{5} \cos \frac{3\pi}{5}$$

$$-1 = 4 \cos \frac{\pi}{5} \cos \frac{3\pi}{5}$$

$$-\frac{1}{4} = \cos \frac{\pi}{5} \cos \frac{3\pi}{5}$$

Marks	Criteria
3	Provides correct solution
2	Provides two different factorisations of $z^5 + 1 = 0$ and equates them, or equivalent merit
1	Provides one correct factorisation of $z^5 + 1 = 0$, or equivalent merit

- b) The polynomial $P(z) = z^4 - 4z^3 + 14z^2 - 36z + 45$ has a purely imaginary root. 3
Find all the roots of $P(z) = 0$.

Sample solution:

Complex roots appear in conjugate pairs, so let the roots be $a + bi$, $a - bi$, ki , $-ki$

Sum of roots:

$$4 = a + bi + a - bi + ki - ki$$

$$4 = 2a$$

$$a = 2$$

Roots 3 at a time:

$$36 = (2 + bi)(2 - bi)(ki) + (2 + bi)(2 - bi)(-ki) + (2 + bi)(ki)(-ki) + (2 - bi)(ki)(-ki)$$

$$36 = 2k^2 + k^2bi + 2k^2 - k^2bi$$

$$36 = 4k^2$$

$$k^2 = 9$$

$$k = \pm 3$$

Product of roots:

$$45 = (2 + bi)(2 - bi)(3i)(-3i)$$

$$45 = 9(4 + b^2)$$

$$5 = 4 + b^2$$

$$b^2 = 1$$

$$b = \pm 1$$

Hence the roots are $3i$, $-3i$, $2 + i$, $2 - i$

Marks	Criteria
3	Provides correct solution
2	Finds two of the three required coefficients, or finds the imaginary roots and develops the equation $z^2 - 4z + 5$ or equivalent merit
1	Defines the four roots and forms one correct equation involving sums/products of roots, or finds the imaginary root and its conjugate or equivalent merit

- c) Let a , b , c be the side lengths of a triangle. 3

Prove that the following inequality must hold:

$$\frac{a}{b+c-a} + \frac{b}{c+a-b} + \frac{c}{a+b-c} \geq 3$$

Sample solution:

By triangle inequality:

$$a + b > c \Rightarrow a + b - c > 0$$

$$b + c > a \Rightarrow b + c - a > 0$$

$$a + c > b \Rightarrow a + c - b > 0$$

Let:

$$x = b + c - a > 0$$

$$y = c + a - b > 0$$

$$z = a + b - c > 0$$

Then:

$$x + y = 2c \Rightarrow c = \frac{1}{2}(x + y)$$

$$x + z = 2b \Rightarrow b = \frac{1}{2}(x + z)$$

$$y + z = 2a \Rightarrow a = \frac{1}{2}(y + z)$$

RTP:

$$\frac{a}{b+c-a} + \frac{b}{c+a-b} + \frac{c}{a+b-c} \geq 3$$

$$\begin{aligned} LHS &= \frac{\frac{1}{2}(y+z)}{x} + \frac{\frac{1}{2}(x+z)}{y} + \frac{\frac{1}{2}(x+y)}{z} \\ &= \frac{1}{2} \left(\left(\frac{x}{y} + \frac{y}{x} \right) + \left(\frac{x}{z} + \frac{z}{x} \right) + \left(\frac{y}{z} + \frac{z}{y} \right) \right) \end{aligned}$$

By AM/GM inequality:

$$\frac{x}{y} + \frac{y}{x} \geq 2 \sqrt{\frac{x}{y} \times \frac{y}{x}}$$

$$\frac{x}{y} + \frac{y}{x} \geq 2$$

Hence:

$$\begin{aligned} LHS &= \frac{1}{2} \left(\left(\frac{x}{y} + \frac{y}{x} \right) + \left(\frac{x}{z} + \frac{z}{x} \right) + \left(\frac{y}{z} + \frac{z}{y} \right) \right) \\ &\geq \frac{1}{2} (2 + 2 + 2) \\ &\geq 3 \text{ as required} \end{aligned}$$

Marks	Criteria
3	Provides correct proof
2	Makes significant progress towards correct solution
1	Correctly applies triangle inequality or AM/GM inequality at an appropriate point in proof, or equivalent merit

d) (i) Show that $\int_0^a f(x) dx = \int_0^a f(a-x) dx$.

2

Sample solution:

Consider $\int_0^a f(u) du$

Let $u = a - x$

$$\frac{du}{dx} = -1$$

$$du = -dx$$

$$u = 0 \Rightarrow x = a$$

$$u = a \Rightarrow x = 0$$

$$\begin{aligned}\int_0^a f(x) dx &= \int_0^a f(u) du \\ &= \int_a^0 f(a-x)(-dx) \\ &= \int_0^a f(a-x) dx\end{aligned}$$

Marks	Criteria
2	Provides correct solution
1	Correctly finds new bounds of integration and finds $\frac{du}{dx}$ in order to implement integration by substitution, or equivalent merit

(iii) Hence evaluate $\int_0^{\frac{\pi}{2}} \frac{1}{1+\tan(x)} dx$.

2

Sample solution:

$$\begin{aligned}
 I &= \int_0^{\frac{\pi}{2}} \frac{1}{1+\tan(x)} dx \\
 &= \int_0^{\frac{\pi}{2}} \frac{1}{1+\tan\left(\frac{\pi}{2}-x\right)} dx \\
 &= \int_0^{\frac{\pi}{2}} \frac{1}{1+\cot(x)} dx \\
 &= \int_0^{\frac{\pi}{2}} \frac{\tan(x)}{\tan(x)+1} dx \\
 2I &= \int_0^{\frac{\pi}{2}} \frac{1}{1+\tan(x)} dx + \int_0^{\frac{\pi}{2}} \frac{\tan(x)}{1+\tan(x)} dx \\
 &= \int_0^{\frac{\pi}{2}} \frac{1+\tan(x)}{1+\tan(x)} dx \\
 I &= \frac{1}{2} \int_0^{\frac{\pi}{2}} 1 dx \\
 &= \frac{1}{2} [x]_0^{\frac{\pi}{2}} \\
 &= \frac{1}{2} \left(\frac{\pi}{2} - 0 \right) \\
 &= \frac{\pi}{4}
 \end{aligned}$$

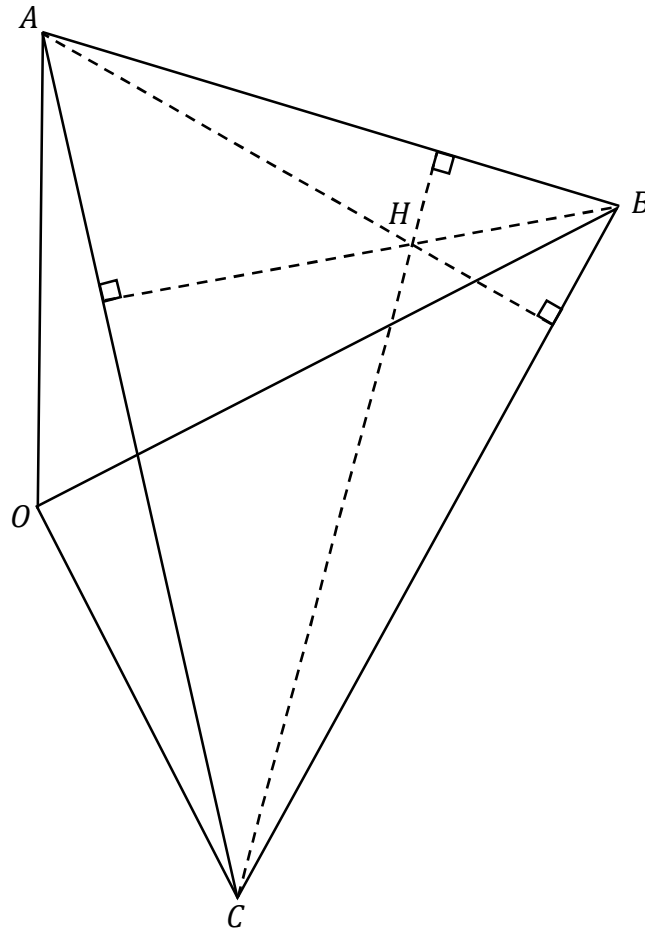
Marks	Criteria
2	Provides correct solution
1	Correctly uses result from part (i) and uses complementary angle results and reciprocal ratios to form a different expression for the integral in terms of $\tan x$, or equivalent merit

Question 16 (15 marks)

- a) In the tetrahedron $OABC$, the edges OA , OB and OC are mutually perpendicular.
Let $\overrightarrow{OA} = \underline{\underline{a}}$, $\overrightarrow{OB} = \underline{\underline{b}}$, $\overrightarrow{OC} = \underline{\underline{c}}$.

4

The altitude of a triangle is the perpendicular line segment drawn from any vertex to the opposite side. Every triangle has three altitudes, which intersect at a single point called the orthocentre. Let H be the orthocentre of triangle ABC .



Similarly, the altitude of a pyramid is the perpendicular line segment drawn from any vertex to the opposite face. Consider the altitude of pyramid $OABC$, drawn from O and perpendicular to triangle ABC . Use vector methods to show that this altitude passes through H .

Sample solution 1:

Let $\overrightarrow{OH} = \underline{h}$

RTP: $\overrightarrow{OH}$ is perpendicular to ΔABC

i.e. need to show that $OH \perp AB, OH \perp AC, OH \perp BC$

Since OA, OB and OC are mutually perpendicular, $\underline{a} \cdot \underline{b} = \underline{b} \cdot \underline{c} = \underline{a} \cdot \underline{c} = 0$

Since H lies on the altitudes of ΔABC , $\overrightarrow{AH} \cdot \overrightarrow{BC} = \overrightarrow{BH} \cdot \overrightarrow{AC} = \overrightarrow{CH} \cdot \overrightarrow{AB} = 0$

$$\begin{aligned}\overrightarrow{AH} \cdot \overrightarrow{BC} &= (\underline{h} - \underline{a}) \cdot (\underline{c} - \underline{b}) \\ 0 &= \underline{h} \cdot \underline{c} - \underline{h} \cdot \underline{b} - \underline{a} \cdot \underline{c} + \underline{a} \cdot \underline{b} \\ 0 &= \underline{h} \cdot \underline{c} - \underline{h} \cdot \underline{b} \\ \underline{h} \cdot \underline{c} &= \underline{h} \cdot \underline{b}\end{aligned}$$

$$\begin{aligned}\overrightarrow{BH} \cdot \overrightarrow{AC} &= (\underline{h} - \underline{b}) \cdot (\underline{c} - \underline{a}) \\ 0 &= \underline{h} \cdot \underline{c} - \underline{h} \cdot \underline{a} - \underline{b} \cdot \underline{c} + \underline{b} \cdot \underline{a} \\ 0 &= \underline{h} \cdot \underline{c} - \underline{h} \cdot \underline{a} \\ \underline{h} \cdot \underline{c} &= \underline{h} \cdot \underline{a}\end{aligned}$$

$$\begin{aligned}\overrightarrow{CH} \cdot \overrightarrow{AB} &= (\underline{h} - \underline{c}) \cdot (\underline{a} - \underline{b}) \\ 0 &= \underline{h} \cdot \underline{a} - \underline{h} \cdot \underline{b} - \underline{c} \cdot \underline{a} + \underline{c} \cdot \underline{b} \\ 0 &= \underline{h} \cdot \underline{a} - \underline{h} \cdot \underline{b} \\ \underline{h} \cdot \underline{a} &= \underline{h} \cdot \underline{b}\end{aligned}$$

Hence $\underline{h} \cdot \underline{a} = \underline{h} \cdot \underline{b} = \underline{h} \cdot \underline{c} = 0$

$$\begin{array}{lll}\overrightarrow{OH} \cdot \overrightarrow{AB} = \underline{h} \cdot (\underline{b} - \underline{a}) & \overrightarrow{OH} \cdot \overrightarrow{BC} = \underline{h} \cdot (\underline{c} - \underline{b}) & \overrightarrow{OH} \cdot \overrightarrow{AC} = \underline{h} \cdot (\underline{c} - \underline{a}) \\ = \underline{h} \cdot \underline{b} - \underline{h} \cdot \underline{a} & = \underline{h} \cdot \underline{c} - \underline{h} \cdot \underline{b} & = \underline{h} \cdot \underline{c} - \underline{h} \cdot \underline{a} \\ = 0 & = 0 & = 0\end{array}$$

Hence $\overrightarrow{OH} \perp \Delta ABC$, so the altitude from O passes through H .

Sample solution 2:

Let P be the foot of the perpendicular from O to ΔABC . Let $\overrightarrow{OP} = \underset{\sim}{p}$.

RTP: Point P is point H

i.e. need to show that $AP \perp BC$, $BP \perp AC$, $CP \perp AB$

Since OA , OB and OC are mutually perpendicular, $\underset{\sim}{a} \cdot \underset{\sim}{b} = \underset{\sim}{b} \cdot \underset{\sim}{c} = \underset{\sim}{a} \cdot \underset{\sim}{c} = 0$

Since $\overrightarrow{OP}$ is perpendicular to ΔABC , $\underset{\sim}{p} \cdot \overrightarrow{AB} = \underset{\sim}{p} \cdot \overrightarrow{AC} = \underset{\sim}{p} \cdot \overrightarrow{BC} = 0$

$$\begin{array}{lll} \underset{\sim}{p} \cdot \overrightarrow{AB} = 0 & \underset{\sim}{p} \cdot \overrightarrow{AC} = 0 & \underset{\sim}{p} \cdot \overrightarrow{BC} = 0 \\ \underset{\sim}{p} \cdot (\underset{\sim}{b} - \underset{\sim}{a}) = 0 & \underset{\sim}{p} \cdot (\underset{\sim}{c} - \underset{\sim}{a}) = 0 & \underset{\sim}{p} \cdot (\underset{\sim}{c} - \underset{\sim}{b}) = 0 \\ \underset{\sim}{p} \cdot \underset{\sim}{b} - \underset{\sim}{p} \cdot \underset{\sim}{a} = 0 & \underset{\sim}{p} \cdot \underset{\sim}{c} - \underset{\sim}{p} \cdot \underset{\sim}{a} = 0 & \underset{\sim}{p} \cdot \underset{\sim}{c} - \underset{\sim}{p} \cdot \underset{\sim}{b} = 0 \end{array}$$

Hence $\underset{\sim}{p} \cdot \underset{\sim}{a} = \underset{\sim}{p} \cdot \underset{\sim}{b} = \underset{\sim}{p} \cdot \underset{\sim}{c}$

$$\begin{aligned} \overrightarrow{AP} \cdot \overrightarrow{BC} &= (\underset{\sim}{p} - \underset{\sim}{a}) \cdot (\underset{\sim}{c} - \underset{\sim}{b}) \\ &= \underset{\sim}{p} \cdot \underset{\sim}{c} - \underset{\sim}{p} \cdot \underset{\sim}{b} - \underset{\sim}{a} \cdot \underset{\sim}{c} + \underset{\sim}{a} \cdot \underset{\sim}{b} \\ &= 0 \end{aligned}$$

$$\begin{aligned} \overrightarrow{BP} \cdot \overrightarrow{AC} &= (\underset{\sim}{p} - \underset{\sim}{b}) \cdot (\underset{\sim}{c} - \underset{\sim}{a}) \\ &= \underset{\sim}{p} \cdot \underset{\sim}{c} - \underset{\sim}{p} \cdot \underset{\sim}{a} - \underset{\sim}{b} \cdot \underset{\sim}{c} + \underset{\sim}{b} \cdot \underset{\sim}{a} \\ &= 0 \end{aligned}$$

$$\begin{aligned} \overrightarrow{CP} \cdot \overrightarrow{AB} &= (\underset{\sim}{p} - \underset{\sim}{c}) \cdot (\underset{\sim}{b} - \underset{\sim}{a}) \\ &= \underset{\sim}{p} \cdot \underset{\sim}{b} - \underset{\sim}{p} \cdot \underset{\sim}{a} - \underset{\sim}{c} \cdot \underset{\sim}{b} + \underset{\sim}{c} \cdot \underset{\sim}{a} \\ &= 0 \end{aligned}$$

Since P lies on all three altitudes of ΔABC , point P must be point H .

Marks	Criteria
4	Provides correct solution
3	Shows that P lies on at least one of the three altitudes of ΔABC , or equivalent merit
2	Shows that $\vec{h} \cdot \vec{a} = \vec{h} \cdot \vec{b} = \vec{h} \cdot \vec{c}$ OR shows that $\vec{p} \cdot \vec{a} = \vec{p} \cdot \vec{b} = \vec{p} \cdot \vec{c}$ (method 2), or equivalent merit
1	Makes a clear statement about what property is required to be proved (i.e. relevant dot product equal to zero) OR states that $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{c} = \vec{a} \cdot \vec{c} = 0$ OR states that $\vec{AH} \cdot \vec{BC} = \vec{BH} \cdot \vec{AC} = \vec{CH} \cdot \vec{AB} = 0$, or equivalent merit

b) Find $\int \frac{1}{(x^2+1)^2} dx$.

3

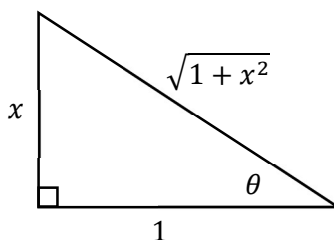
Sample solution:

Let:

$$x = \tan \theta$$

$$\frac{dx}{d\theta} = \sec^2 \theta$$

$$dx = \sec^2 \theta d\theta$$



$$\begin{aligned} \int \frac{1}{(x^2+1)^2} dx &= \int \frac{1}{(1+\tan^2 \theta)^2} \sec^2 \theta d\theta \\ &= \int \frac{1}{\sec^4 \theta} \sec^2 \theta d\theta \\ &= \int \cos^2 \theta d\theta \\ &= \frac{1}{2} \int 1 + \cos 2\theta d\theta \\ &= \frac{1}{2} \left(\theta + \frac{1}{2} \sin 2\theta \right) + c \\ &= \frac{\theta}{2} + \frac{1}{2} \sin \theta \cos \theta + c \\ &= \frac{1}{2} \tan^{-1} x + \frac{1}{2} \times \frac{x}{\sqrt{1+x^2}} \times \frac{1}{\sqrt{1+x^2}} + c \\ &= \frac{1}{2} \tan^{-1} x + \frac{x}{2(1+x^2)} + c \end{aligned}$$

Marks	Criteria
3	Provides correct solution
2	Finds a primitive in terms of θ
1	Uses substitution to express integral in terms of θ

c) Prove that, for any real numbers a and b , $a^2 + b^2 - ab - a - b + 1 \geq 0$.

2

Sample solution:

$$\begin{aligned} a^2 + b^2 - ab - a - b + 1 &= \frac{1}{2}(2a^2 + 2b^2 - 2ab - 2a - 2b + 2) \\ &= \frac{1}{2}((a^2 - 2ab + b^2) + (a^2 - 2a + 1) + (b^2 - 2b + 1)) \\ &= \frac{1}{2}((a - b)^2 + (a - 1)^2 + (b - 1)^2) \\ &\geq 0 \quad \text{since } (a - b)^2 \geq 0, (a - 1)^2 \geq 0, (b - 1)^2 \geq 0 \end{aligned}$$

Marks	Criteria
2	Provides correct proof
1	Expresses LHS as the sum of perfect squares

d) Consider the integral $I_n = \int_0^1 (1 - x^2)^n dx$ for integers $n \geq 0$.

(i) Show that for $n \geq 1$, the recurrence relation $I_n = \frac{2n}{2n+1} I_{n-1}$ holds.

3

Sample solution:

Let: $u = (1 - x^2)^n$ $v' = 1$

$u' = n(1 - x^2)^{n-1}(-2x)$ $v = x$

$$I_n = [x(1 - x^2)^n]_0^1 - \int_0^1 -2nx^2(1 - x^2)^{n-1} dx$$

$$= 0 - 2n \int_0^1 -x^2(1 - x^2)^{n-1} dx$$

$$= -2n \int_0^1 (1 - x^2 - 1)(1 - x^2)^{n-1} dx$$

$$= -2n \int_0^1 (1 - x^2)^n - (1 - x^2)^{n-1} dx$$

$$I_n = -2nI_n + 2nI_{n-1}$$

$$(1 + 2n)I_n = 2nI_{n-1}$$

$$I_n = \frac{2n}{2n+1} I_{n-1}$$

Marks	Criteria
3	Provides correct solution
2	Correctly uses integration by parts to obtain a single integral term, or equivalent merit
1	Attempts to use integration by parts, or equivalent merit

(ii) Hence, show that $I_n = \frac{4^n(n!)^2}{(2n+1)!}$.

3

Sample solution:

$$\begin{array}{llll}
 I_0 = \int_0^1 1 \, dx & I_1 = \frac{2}{3} I_0 & I_2 = \frac{4}{5} I_1 & I_3 = \frac{6}{7} I_2 \\
 = [x]_0^1 & = \frac{2}{3} \times 1 & = \frac{4}{5} \times \frac{2}{3} & = \frac{6}{7} \times \frac{4}{5} \times \frac{2}{3} \\
 = 1 & = \frac{2}{3} & &
 \end{array}$$

$$\begin{aligned}
 I_n &= \underbrace{\frac{2n}{2n+1} \times \frac{2(n-1)}{2(n-1)+1} \times \frac{2(n-2)}{2(n-2)+1} \times \cdots \times \frac{6}{7} \times \frac{4}{5} \times \frac{2}{3}}_{n \text{ terms}} \\
 &= \frac{2^n(n(n-1)(n-2) \cdots \times 3 \times 2 \times 1)}{(2n+1)(2n-1)(2n-3) \times 7 \times 5 \times 3} \\
 &= \frac{2^n(n!)}{(2n+1)!} \times \underbrace{(2n(2n-2)(2n-4) \cdots \times 6 \times 4 \times 2)}_{n \text{ terms}} \\
 &= \frac{2^n(n!)}{(2n+1)!} \times (2^n(n!)) \\
 &= \frac{2^{2n}(n!)^2}{(2n+1)!} \\
 &= \frac{4^n(n!)^2}{(2n+1)!}
 \end{aligned}$$

Marks	Criteria
3	Provides correct solution
2	Develops an expanded expression for I_n
1	Finds values of I_0 , I_1 and I_2

End of paper